Pale Bands in the Affine Heatmaps: What They Are, and the Search for Hidden Computation

local_mixing working notes

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Abstract

Reading the layer-2 profile sweep's heatmaps at pixel resolution turned up narrow vertical bands where the mixed circuit is markedly less affine-readable. This note establishes what they are. Three results, in order of importance. (1) The heatmap measure carries *no sampling noise*: every cell is an exact count of non-recoverable target bits, and the plate is byte-identical under different random seeds — so every visible feature is real structure, and the "is it jitter" question is answered before it is asked. (2) The bands are genuine local regions, 200–2200 gates wide, where 40–75 more of the 512 target bits go unreadable. (3) They are *dead*, not *hidden*: a whole-circuit search across every position and every gap width up to 11k gates finds no case where the computation re-emerges further along, with the observed statistic sitting far *below* its own null. The interesting phenomenon — mixed material performing part of the source computation invisibly — does not occur in this mixer's output at any measurable resolution.

1 The instrument has no noise floor

`hmap_affine` scores cell (i, j) by fitting, for each of the n target bits of C's prefix i, a GF(2) affine function of the mixed circuit's wires at prefix j: inconsistent (the bit is not affine in those wires) scores 0.5; consistent scores the measured holdout error, which is ≈ 0 for a genuine affine relation. So

$$H(i, j) \;=\; \tfrac{1}{2} \cdot \big(\text{fraction of } C\text{'s bits not affine-recoverable from } G_j\big).$$

Two checks confirm this is exact rather than statistical:

- **Quantisation.** Every cell of the 101×121 `nR30` plate is an exact integer multiple of $0.5/n = 0.5/512 = 0.000977$ — 100% of cells on the lattice, to within 10^{-3} of a multiple.
- **Seed independence.** Runs at seeds 12345, 999, 4242 and 31337 produce **byte-identical** `.bin` files, as does a minimal control at seeds 7 and 88888. Span membership over GF(2) is a fact about the functions, not about which random inputs were drawn.

Consequence: there is no noise to discount. A column that looks pale *is* pale. (An earlier "how many standard errors" framing in this investigation was therefore mis-conceived, and is superseded.)

Caveat worth carrying: because the plate is seed-independent, one cannot use re-seeding to cross-check a feature, and agreement between the degree-1 and degree-2 plates is **not** independent evidence — both draw their sample batches from the same seeded stream, so the degree-2 run's first 96 batches are byte-identical to the degree-1 run's.

2 What the bands are

2.1 Width and depth

On the coarse plate (column spacing 1876 gates) 17 of 121 columns sit more than 2 local units above baseline. Re-measured at 200-gate spacing (26×1127), the features resolve into structured bands with smooth shoulders:

G-gate range	width (gates)	peak H	peak non-recoverable bits
13,200–15,200	2,200	0.3941	**404** of 512
97,600–98,800	1,400	0.3805	390
178,200–178,800	800	0.3837	393
20,800–21,400	800	0.3853	395
125,200–125,600	600	0.3622	371
four more	200–400	0.367–0.372	377–381
typical material	—	0.3226	330

Table 1: `nR30`. The ordinary column-to-column wobble on this plate is 7.5 bits of 512; the bands stand well clear of it.

2.2 They appear in every arm

Each of the eight profile arms has them; `nR30` merely has the most, and its larger circuit spaces them far enough apart to be seen separately. Per-arm strongest band (excluding edge columns), all rendered at matched aspect and a common colour scale:

arm	size ratio	band G	peak H	bits
no twist	1.30	40,145	0.3535	362
R1=1.5	1.49	6,230	0.3611	370
R1=2.0	1.79	137,700	0.3605	369
R1=2.5	1.99	6,648	0.3808	390
R1=3.0	2.25	97,552	0.3684	377
short hold	1.62	37,800	0.3501	359
long hold	1.98	108,636	0.3467	355
10× twist	6.42	176,484	0.3632	372

2.3 Framing note: the plate's aspect is the size ratio

The ridge's slope is $\mathrm{d}G/\mathrm{d}C = |G|/|C|$. For `nR30` that is $225{,}014/100{,}000 = 2.25$, and the measured ridge slope is 2.183 (fractional slope 0.970) — the diagonal crosses corner to corner, as it must. A local window therefore needs its G extent to be $\approx 2.25\times$ its C extent or the ridge sweeps out of frame; the first round of zoom plates in this investigation was mis-framed for exactly this reason. The per-row ratio also drifts from 2.63 early to 2.25 late, i.e. material that was present early is proportionally more inflated — a residue of the expansion phase that compression did not re-flatten.

3 Dead, not hidden

3.1 The question

A pale band where the ridge resumes at the *same* C position is uninteresting: it is a stretch that does no work (an identity, in effect). The interesting event would be a band after which the computation **re-emerges further along** — mixed material that performed part of C invisibly and handed back the result. That is a specific, testable signature: a persistent forward level-shift in the ridge trajectory $C^*(G)$, beyond what the size ratio predicts.

3.2 The search

Two whole-circuit plates at 501×641 cells (C step 200 gates, G step 280–352), so the detector scans *every* position rather than a hand-picked window. For each candidate position the trend fitted on the left of the gap is extrapolated across it and compared with the level on the right; the gap width is swept from 1 to 32 columns (352 up to 11,264 gates). The null detrends the true ridge and *shuffles its residuals*, preserving the wobble's magnitude while destroying its structure, then runs the identical detector.

circuit	gap (gates)	observed	null p95	null max	p
nR30	352	1,125	4,418	4,837	1.000
nR30	2,816	2,525	7,518	8,273	1.000
nR30	11,264	5,783	17,764	19,677	1.000
nR20	280	875	2,298	2,888	1.000
nR20	2,240	2,075	3,916	4,529	1.000
nR20	8,960	6,175	9,624	11,376	0.950

Table 2: Largest persistent forward shift in C gates. Every cell $p \geq 0.95$.

3.3 Verdict

No re-emergence events, at any scale, anywhere in either circuit. The result is stronger than a bare null: the observed statistic sits *far below* its own null distribution, typically a quarter to a half. Shuffling destroys the autocorrelation of the ridge's wobble and manufactures apparent leaps that the true trajectory never makes — the real ridge is **smoother than chance**. C's computation advances steadily at the size-ratio rate; the bands are stretches where it does not advance and then resumes exactly where it left off.

An earlier pass of this analysis, restricted to eight 6000-gate windows, appeared to find persistent shifts of 430–820 C-gates in all eight arms. That was the maximum of a noisy statistic over $\sim$120 candidate positions, and the null test dissolved it ($p = 0.05$ to 1.00). It is recorded here because the failure mode — cherry-picking an extremum without a null — is the one this kind of search invites.

4 Incidental findings

- **Ridge scatter scales with expansion.** `nR30`'s ridge wobbles by 880 C-gates about its trend, `nR20`'s by 359. The more expanded circuit localises its computational progress less tightly — a mild point in expansion's favour that the summary statistics miss.

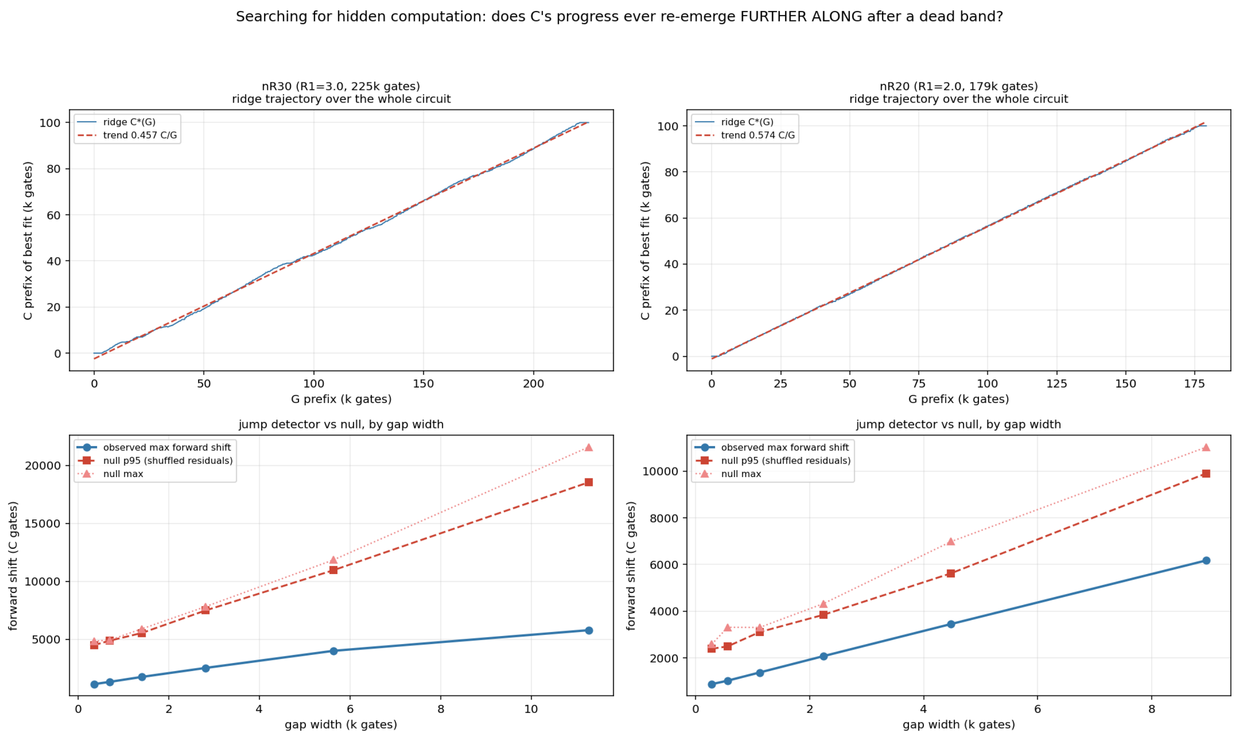


Figure 1: Top: the ridge trajectory over the whole circuit against its fitted trend. Bottom: observed maximum forward shift against the shuffled-residual null, by gap width. The observed curve lies below the null everywhere.

- **Mean H rises with circuit size** (0.337 at 130k gates to 0.361 at 642k) while ridge *depth* stays flat at 0.174–0.188. The drift is dilution, not a weakening diagonal: per the standing directive, read the plate by ridge, never by the mean.
- **Two arms' strongest band sits in the first few percent** of the circuit (R1=1.5, R1=2.5). Plausibly a head effect distinct from the mid-circuit bands; not investigated.

5 Tools added

- `reports/hmap_pixels.py` — renders a plate at true 1:1, one image pixel per matrix cell, with `-scale` for integer magnification, `-gates A:B` to crop by G-gate index and `-annotate` to print the numbers behind the pixels. Matplotlib's figure rendering resamples plates and smears single-column features into their neighbours; this does not.
- `hmap_affine -c-from/-c-to/-g-from/-g-to` — restricts the plate to a rectangle of prefix indices, so a dense *local* plate costs minutes instead of computing a dense global one.
- `plot_hmap_ridge.py -no-ridge` — draws the raw H field without the traced overlay, while still computing and printing the ridge statistics.

A The plates

Every image below is rendered by `hmap_pixels.py`: one image pixel per matrix cell, integer replication only, no interpolation and no resampling. Colour is fixed across the whole appendix — $H = 0.10$ (red, leaking) to $H = 0.45$ (blue, hidden) — so panels are directly comparable. In every plate rows are C prefixes (top = start of the source computation) and columns are G prefixes (left = start of the mixed circuit).

A.1 A resolution ladder on one circuit

All five panels show the same object — `nR30`, 225,014 gates against a 100,000-gate source — sampled at five resolutions. The point of the ladder is that what a plate *says* depends on how finely it is cut: a coarse plate shows a clean diagonal and a handful of suspicious columns; a fine one shows those columns are structured bands with shoulders; a local one shows the diagonal itself has width and texture.

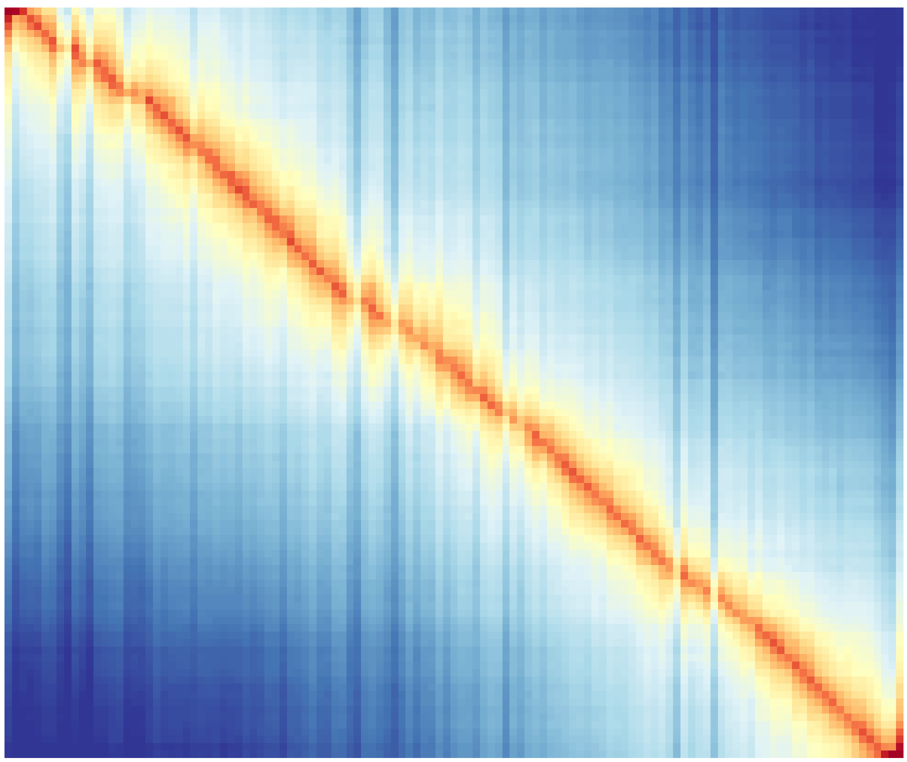


Figure 2: **Rung 1 — global, coarse.** 101×121 cells; C step 1000, G step 1876 gates. This is the plate the sweep was read from. The diagonal is unmistakable; the pale columns are visible but unresolved, each one cell wide.

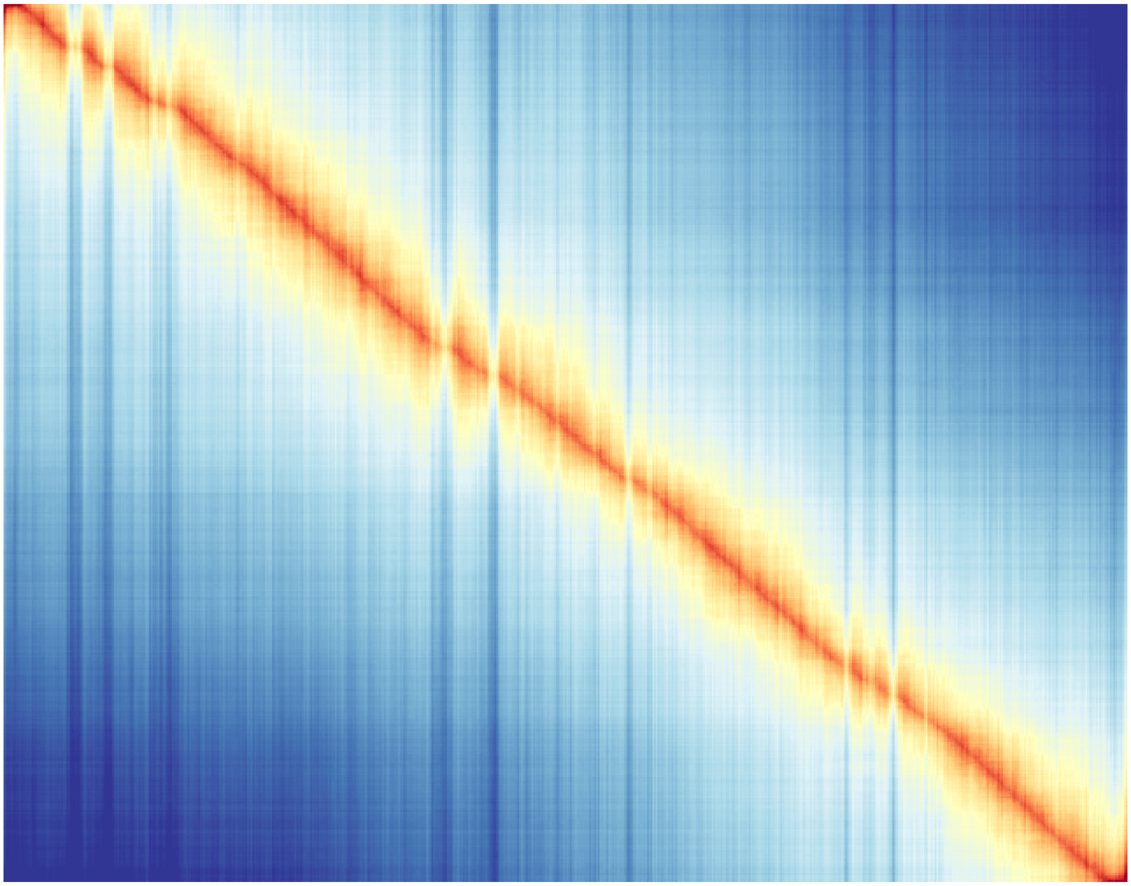


Figure 3: **Rung 2 — global, dense in C.** 501×641 cells; C step 200, G step 352 gates, one pixel per cell at true 1:1. This is the plate the hidden-computation search ran on: dense enough in C that a forward jump of a few hundred gates would be visible, over the whole circuit rather than a window.

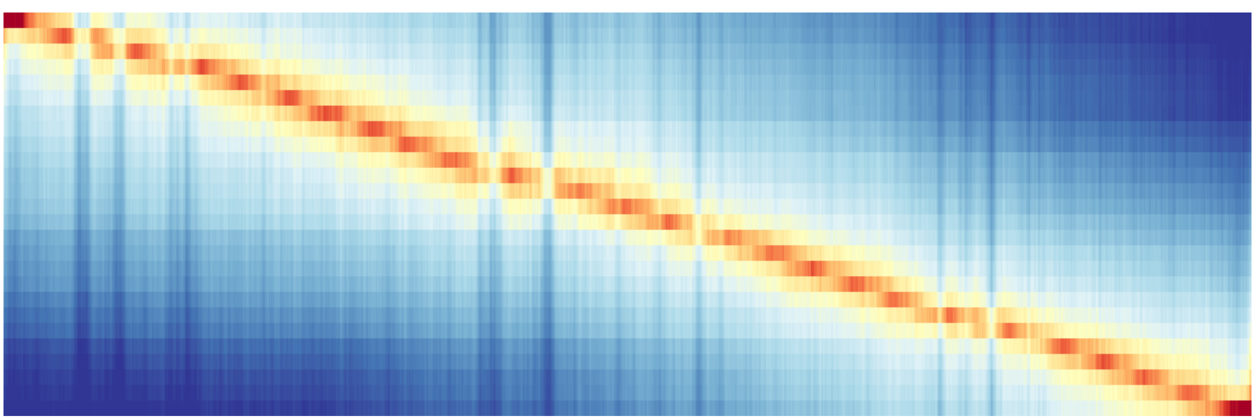


Figure 4: **Rung 3 — fine in G, coarse in C.** 26×1127 cells; G step 200 gates, C step 4000, stretched $14\times$ vertically (pixel replication, not interpolation). This is the cut that resolved the band widths: the pale features are 200–2200 gates across with smooth shoulders, not knife edges.

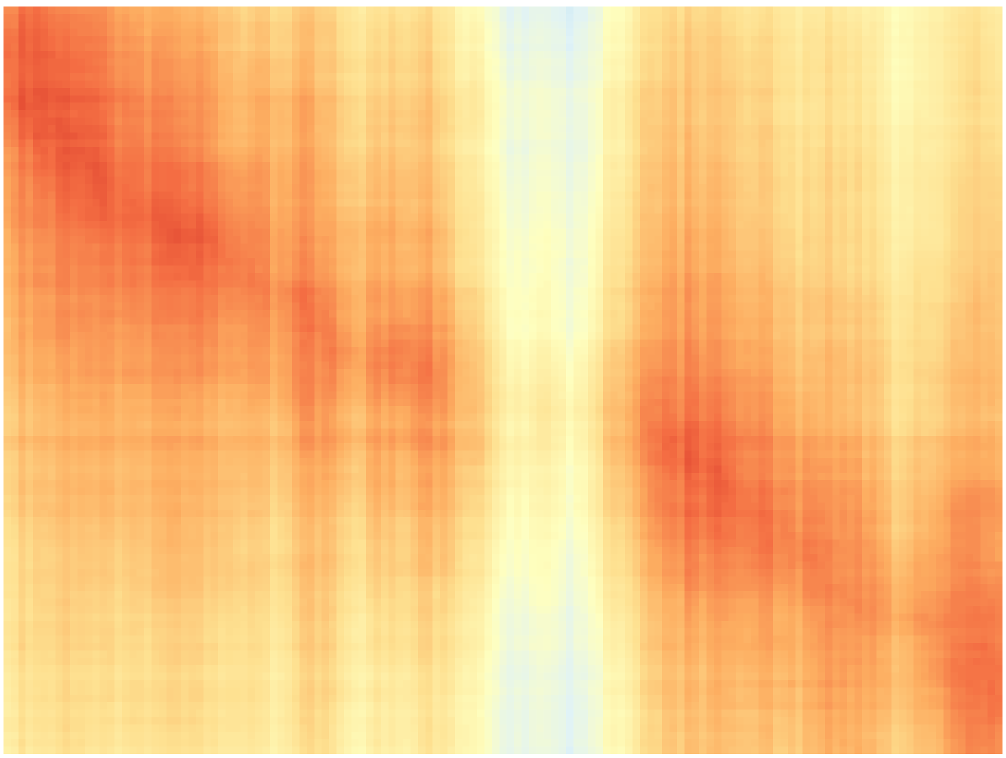


Figure 5: **Rung 4 — local, dense in both.** 101×135 cells; C step 60, G step 100 gates, over $C \in [39\mathrm{k}, 45\mathrm{k}]$, $G \in [90.9\mathrm{k}, 104.3\mathrm{k}]$. At this scale the diagonal is a textured structure of finite width rather than a line.

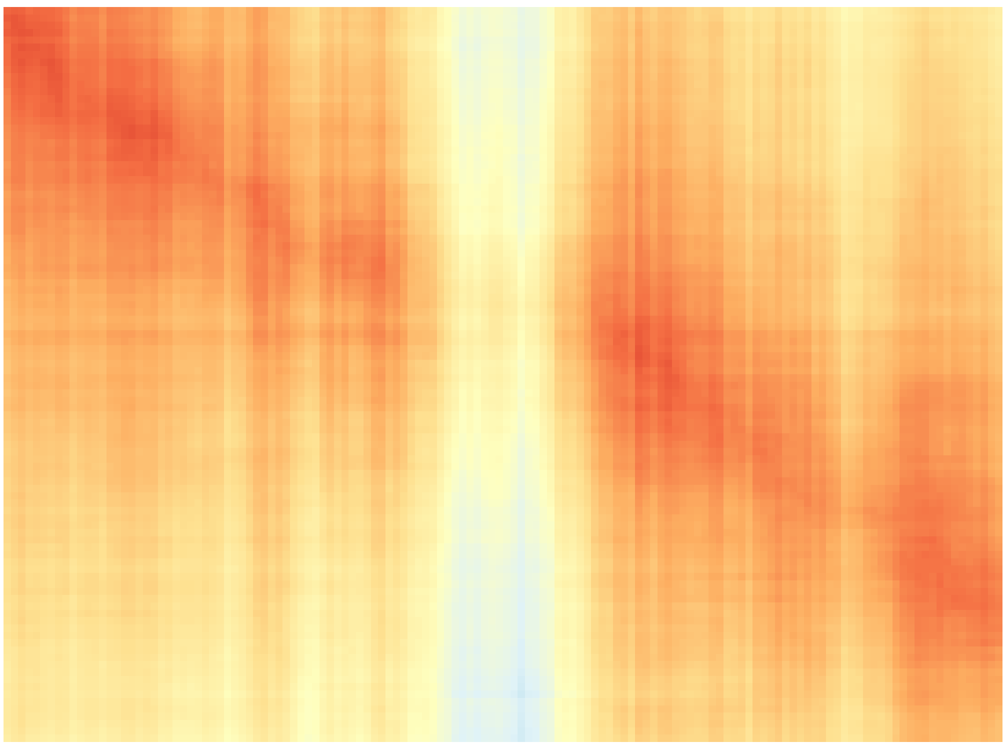
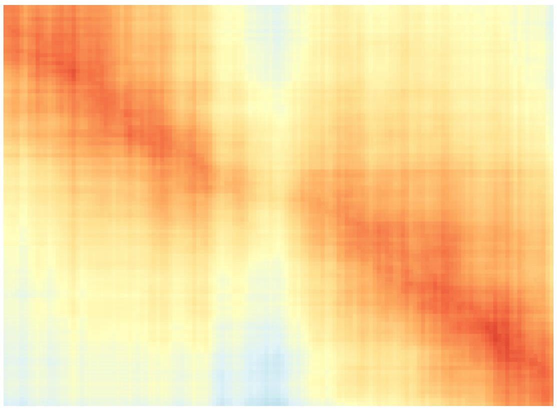


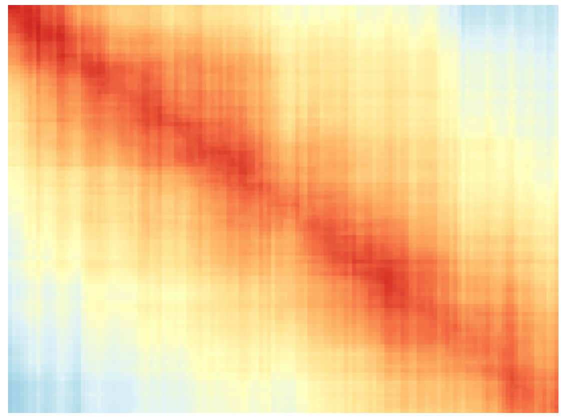
Figure 6: **Rung 5 — local, aspect-matched.** 100×136 cells over C 6000 gates $\times$ G 13,500 gates, i.e. a window whose sides are in the circuit's own 2.25 size ratio, so the ridge runs corner to corner instead of sweeping out of frame. Compare with rung 4: the same material, framed correctly.

A.2 One band from each profile arm

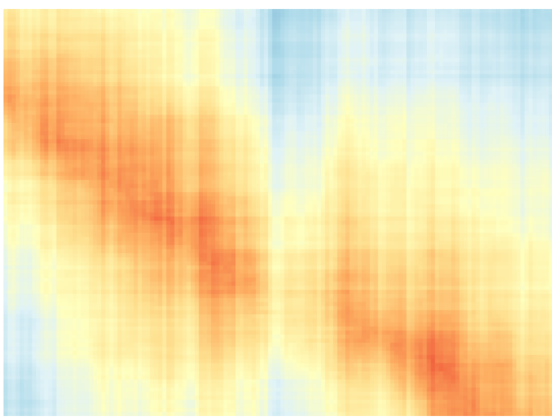
Each panel is that arm's own strongest band, in a window aspect-matched to that arm's own size ratio, at C step 60 gates. Same colour scale throughout (0.10–0.35 here, since these local windows never reach the global extremes). The bands look alike across very different runs — the 6.4×-expanded twist-heavy circuit included — which is the visual counterpart of the finding that no profile parameter changes the plate.



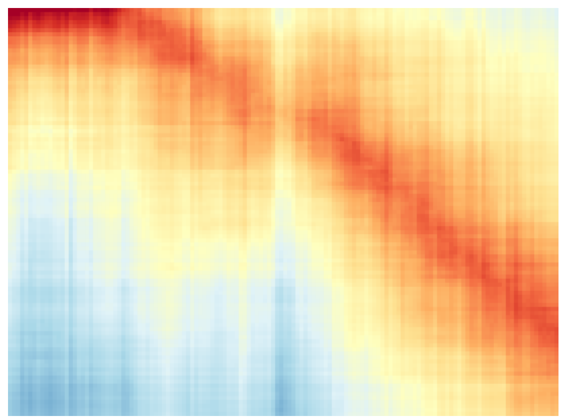
no twist (130k gates, ratio 1.30)



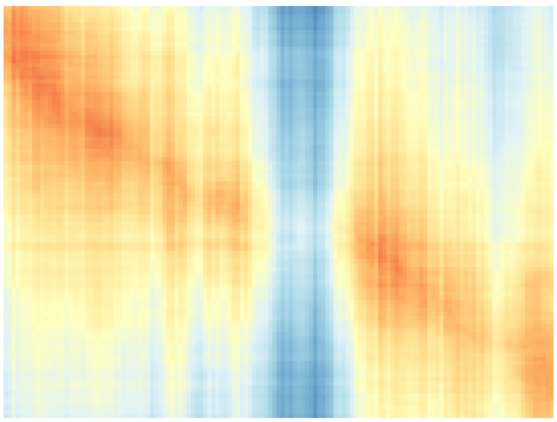
R1=1.5 (149k, 1.49)



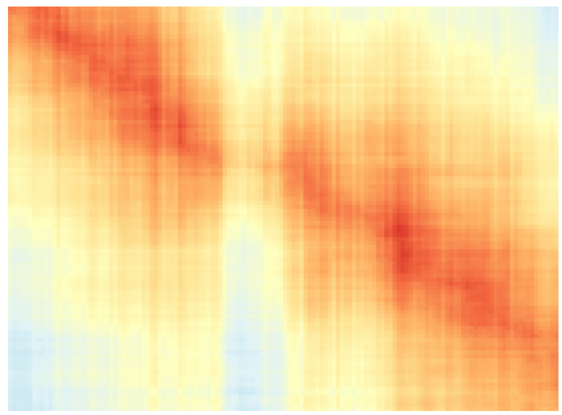
R1=2.0 (179k, 1.79)



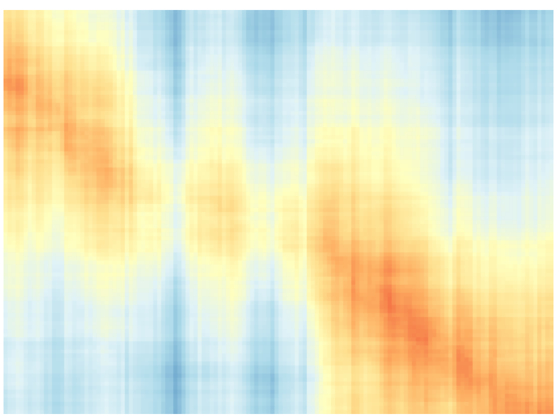
R1=2.5 (199k, 1.99)



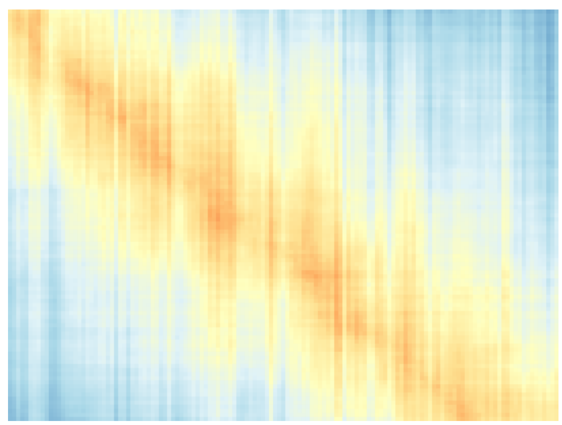
R1=3.0 (225k, 2.25)



short hold (162k, 1.62)



long hold (198k, 1.98)



10× twist (642k, 6.42)